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Large structure in a turbulent boundary layer

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Using an array of hot wires and wall shear stress probes results are presented which support the hypothesis that an organized structure exists in the boundary layer. From the correlation between wall shear stress and velocity it is shown that this structure is at an oblique angle to the free stream. The inferred angle from this correlation is found to agree remarkably with recent flow visualization photographs. Evidence is found that the large scale motion in the organized structure produces a slowly varying component in the wall shear stress and, importantly, a high frequency large amplitude fluctuation occurring near the maximum in the slowly varying wall shear. It is suggested that this latter fluctuation is associated with the bursting phenomenon. A model which accounts for these results is described. It leads to a description of the bursting process in terms of a rotational instability and while too approximate to give more than good order of magnitude estimates, it gives values of the wall streak spacing which are not inconsistent with previous measurements.

I. INTRODUCTION

The structure of the turbulent boundary layer has been the object of an enormous research effort over several decades. No attempt is made here to review this extensive literature; the reader may wish to refer to Willmarth¹ for a recent and excellent review of the progress that has been made. For the present purpose, it is sufficient to note that the structure in the region near the wall, in particular the "bursting" phenomenon (Kline *et al.*² etc.) has been well documented (Willmarth¹ summarizes the many contributions made to build up the present picture), and similarly, the large structure of the outer intermittent flow has been investigated extensively, firstly by space-time correlation techniques and subsequently by conditional sampling techniques (for example, Kaplan and Laufer,³ Kovasznay *et al.*⁴). An essential issue which has remained, however, is the connection between the bursting phenomenon near the wall and the large scale motion in the outer part of the layer. Associated with this is the question of the origin of the bursts and the nature of the cyclical regeneration of bursts. Offen and Kline⁵ have described a model based on flow visualization studies which attempts to provide a partial answer to these questions. They suggest that a circulatory flow, or vortex, in a burst (or lifted wall streak) can impose an adverse pressure gradient on the wall as it passes downstream, and this adverse pressure gradient acting on a newly forming wall streak stimulates the lift-up of this new streak. A part of this hypothesis as suggested by Offen and Kline is that a pairing of the vortices associated with bursts leads to a return of part of an old burst to form the origin of the new burst. The process is thus regenerative. They further suggest that the large scale motion in the outer part of the layer is a consequence of a repeated "pairing" process as the vortices, associated with bursts, are convected into the outer part of the boundary layer. There is an implication perhaps that the outer part of the boundary layer plays a passive role.

While not of themselves sufficient to contradict these hypotheses, there are several results which are not

easily reconciled with them, for example, the important finding of Rao, *et al.*⁶ that the burst rate scales with the outer flow variables. Lu and Willmarth,⁷ working with large spikes in the Reynolds stress, also found a rate which scales with outer flow variables. Similarly, the dramatic effect of drag reducing polymers on the flow near the wall and, in general, the considerable difference in scales, at high Reynolds number, between the initial wall streak (mean spacing approximately $100\nu/U_\tau$) and the motions in the outer region of the boundary layer are not easily incorporated in a hypothesis in which the large scale structure, which is very weakly dependent on Reynolds number, is derived from the amalgamation of bursts near the wall.

A quite different point of view from that of Offen and Kline has been taken by Coles and Barker⁸ in which it is proposed that a large scale structure, in particular, the turbulent spot, is the building block of the turbulent boundary layer and that the bursting phenomenon is part of the signature of this spot. That the bursting phenomenon may be a signature of such a large scale eddy is also implied in what appears, in the light of the present work, to be a far-sighted speculation by Badri Narayanan *et al.*⁹

In addition to the extensive literature on the structure of turbulent boundary layers, the authors have obviously been greatly influenced by earlier work¹⁰ and by continuing studies of the large structure in the mixing layer. In fact, the optimistic aim of this work which has guided the planning and carrying out of these experiments was to try and establish whether a similarly significant organized structure exists in the boundary layer.

As "organized" structure has come to mean different things to different people¹¹ some comment on what is meant by it in the present context is in order. Organized structure is taken here to mean a "recognizable" large scale motion which has its own internal dynamics (a mechanism to generate and transport smaller scale turbulence, for example) and which of itself and through interactions with other similar structures dominates the mechanics of the flow. In some respects this view of

the large structure follows from Townsend's¹² early large eddy hypothesis, but it differs particularly in the way in which the large eddies are formed and the extent to which they are deterministic, in the variety of forms they are expected to take, and correspondingly, in the detailed way in which they dominate the mechanics of the flow, that is, for example, determine the Reynolds stress. (In the results to be discussed, for example, it appears that the Reynolds stress in the mixing layer is essentially determined by the large scale motion of the large structure, i.e., by the two-dimensional motion, whereas a totally different relationship between the large structure and the generation of Reynolds stress is suggested for the boundary layer.)

II. EXPERIMENTAL CONSIDERATIONS

Quite apart from the reality or significance of organized structures in a turbulent flow, what makes any attempt to find them such a difficult problem (fraught with the danger of producing order in chaos as many have recognized) is the randomness in phase between "recognizable" structures, and the "jitter" in the amplitude and the characteristic time scale of these structures. In the boundary layer the problem is made doubly difficult because of the crucial three-dimensionality of the motion and the corresponding randomness in the transverse direction of the slice through any structure obtained at a single measuring station. It is immediately clear, however, that if a large structure exists, there should be some correlation (by definition) between the motion at points throughout the region of the boundary layer which the structure occupies. The experiment was therefore aimed at simultaneous measurements throughout the boundary layer with an array of hot wires and with the intention of using a computer to sort and correlate the data. We were led to build an inexpensive 16 channel analog-to-digital data acquisition system capable of making blocks of 16 000 measurements, at any total rate up to 4 million samples a second and writing them on computer tape. In these experiments this was used to simultaneously sample and record up to seven hot-wire channels.

If, further, there is an organized structure, then it must in some sense be coupled with the bursting phenomenon, but it seemed likely, particularly in view of the somewhat limited results obtained by Blackwelder and Kovasznay¹³ using a burst detecting wire at $y^+ \approx 24$, that the great disparity in scales, the nature of bursting and the general intensity of activity at a y^+ of 24 may make correlations between this region and the outer flow difficult to find. Rather it was felt, more or less intuitively, that the wall shear (ideally, the wall shear measured simultaneously at several points) might be a more suitable indicator of phenomena near the wall with which to correlate measurements in the outer part of the boundary layer. With this in mind extremely small, (0.5 by 0.25 mm) hot-film probes were constructed by the vacuum deposition of nickel on glass. Details of their construction to achieve very accurately repeatable characteristics, the associated high-speed, high-current feed-back bridges used to

operate them, and an analysis of the system time response will be published separately. It is sufficient to note that the response time as measured in a shock tube was less than 50 μ sec, that two separate probes side by side in a boundary layer and operated by separate bridges gave indistinguishable spectra, and when calibrated in a laminar flow in the rotating disc apparatus of Brown and Davey¹⁴ the measurement of mean skin friction under a turbulent boundary layer was within 3% of the value determined from Preston tube measurements. From the calibration of the film the "effective" length of the film (in the terminology of Brown¹⁵) was found to be 2.6 times the measured length (0.25 mm) and in the boundary layer of this study the corresponding value of $U_\tau l_{\text{eff}}/\nu$ was 56. The thermal boundary layer of the film therefore extended only to $y^+ \approx 13$.¹⁵ This, of course, is the thermal boundary layer associated with the mean wall shear; the effective length rapidly reduces to the measured length for fluctuating wall shear¹⁶ so that the thermal boundary layer associated with the fluctuating wall shear is even thinner. Because of the dependence of the effective length on frequency, particularly at low frequencies, the relationship between the fluctuating heat transfer and the wall shear is not simple (of course, for correlation purposes this is not a particular difficulty). Well above a frequency of $\omega l_{\text{eff}}^2/\kappa_{\text{glass}}$, however, where the effective length has become the measured length, the fluctuating wall shear may be straightforwardly obtained from the steady calibration, (taking into account the different effective lengths). [This, of course, assumes quasi-steady heat transfer, which by simple scaling arguments in the governing heat equation is expected to exist for frequencies $\omega\nu/U_\tau^2$ less than $(U_\tau l/\nu)^{-2/3}$ for Prandtl number 1.]

Since the limits in this experiment for these two frequencies are about 50 Hz and 2.5 kHz which correspond to the relative extremes of velocity fluctuation spectra, the output of the film has been converted to a shear stress fluctuation and the results expressed accordingly. Since it is only correlations which are being compared here, however, the results will be seen to be quite independent of this step. Thus, the experimental arrangement was an array of hot wires arranged initially vertically above a wall shear probe in a zero pressure gradient turbulent boundary layer.

The tunnel used is described by Lim.¹⁷ The free stream velocity was 36 m/sec, the boundary layer thickness approximately 40 mm, the displacement thickness 5.51 mm, and at the measuring station the Reynolds number based on momentum thickness was 10 160. The mean velocity profile was entirely standard and the friction velocity determined from a Preston tube was 1.28 m/sec. The aspect ratio of the boundary layer is unavoidably low in this tunnel (tunnel width: boundary layer thickness = 5.8 in the first set of experiments); however, the experiments have been repeated at distances farther upstream in the working section (aspect ratio = 8.5, $Re_\theta = 4,930$) and properly scaled with U_0 and δ there is no apparent difference between the two sets of results.

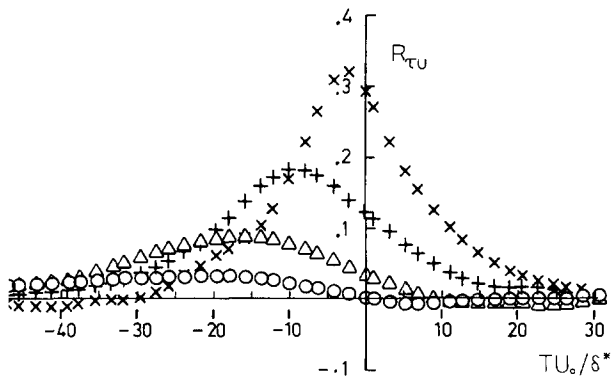


FIG. 1. Long time averaged correlations between the wall shear stress and the velocity at various points directly above the hot film. $\times$, $y/\delta = 0.05$; $+$, $y/\delta = 0.25$; Δ , $y/\delta = 0.50$; $\circ$, $y/\delta = 0.75$.

III. RESULTS

A. Velocity, wall shear correlations

In the first experiment an array of four hot wires was placed immediately above the wall shear probe (at $x=0$) with the wires located at $y/\delta = 0.05, 0.25, 0.5, 0.75$. The long time correlation coefficient defined by

$$R_{\tau u}(T) = \lim_{T_s \rightarrow \infty} \frac{1}{T_s} \int_0^{T_s} \tau(t) u(t+T) dt$$

(where the wall shear τ and the velocity u were non-dimensionalized by their rms values) was determined for the four wires. The results are shown in Fig. 1.

Several points are worth noting. Firstly, there is a quite perceptible correlation between the wall shear and the velocity even at $y = 0.75\delta$. (The absolute magnitude of this correlation is determined by the rms values of τ and u . As will become clear, the high frequency large amplitude wall shear fluctuations contribute more to the rms value than to the correlation so that the magnitude of the correlation appears unnecessarily small.) As remarked in the introduction this correlation at large distances is especially significant given the relatively high Reynolds number; $y = 0.75\delta$ corresponds to approximately 25 wall-streak spacings, for example. Secondly, and not surprisingly, the correlation peaks at increasing time delay as the point of velocity measurement is moved farther from the wall. Both of these results are consistent with measurements reported by Blackwelder and Kovaszny¹³ using a wire at a y^* of 24 instead of a wall shear probe. Their correlations are similar at the same y/δ but extend, interestingly, only to $y/\delta = 0.5$ which by comparison, at their Reynolds number corresponds to only four wall-streak spacings. Perhaps the most important question which these results raise is whether this correlation is due to events at the wall being convected out into the flow along the trajectory implied by the increasing time delay with distance from the wall, or, alternatively, due to some organized structure at an oblique angle to the wall which produces a characteristic response in the wall shear as it passes by. As a first step toward establishing whether either of these views is correct the rake of wires was moved downstream (i.e., chang-

ing x) keeping the same values of y/δ . It was intended to find, experimentally, the points where $R_{\tau u}$ was a maximum at zero time delay ($T=0$). This was quite straightforward. As a first estimate for x it was assumed that the convection speed was $0.7U_0$ and x followed from the values of T at which the $x=0$ correlations peaked. The hot wire was then placed at this value of x , the correlation was found to peak not quite at $T=0$, and the process repeated. Only a few iterations were required. The resulting correlations having a maximum at $T=0$ are shown in Fig. 2. The conclusion that can hardly be avoided on this evidence alone is that of the two hypotheses only the second could account for these results. The correlations which peak at $T=0$ are (within the error bars) practically indistinguishable from those at $x=0$ (with the appropriate shift in time delay). It must be true, given the velocities in a boundary layer, that the probes at this oblique angle are nearer to the trajectory of a burst convected from the wall than when the probes are vertical and if the first hypothesis were correct, then they should give a higher correlation at this oblique angle. Alternatively, one could argue that the same maxima in the correlation but one at zero time delay and the other at a time delay TU_0/δ^* of approximately 25, which is considerably larger than measured correlation time scales in the sublayer,¹⁸ is inconsistent with the first hypothesis.

The location of the probes which gave correlation maxima at $T=0$ is shown in Fig. 3. Evidently, the curve which is a good fit to these four points is very nearly a straight line at an angle of 18° to the wall (free stream). Coincidentally and independently, results from a flow visualization study using smoke were presented at the same meeting as the present results.¹⁹ By courtesy of Dr. Falco a tracing of the outline of the smoke in one of his pictures (selected without knowledge of these results) is shown in Fig. 4. The agreement between the inferred angle obtained by wall shear velocity correlation and an apparent angle in the flow visualization study is quite remarkable.

It is interesting to note that the velocity determined from the time delay of the maxima in the correlation at

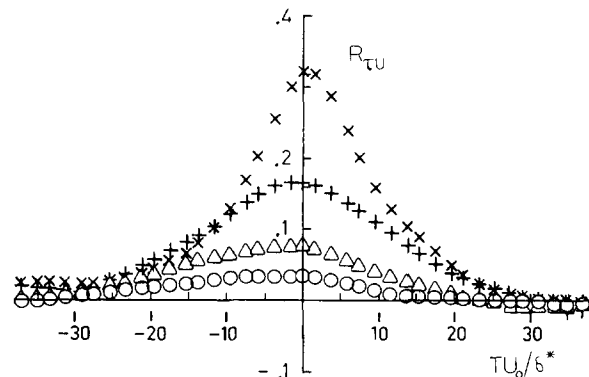


FIG. 2. Long time averaged correlations between the wall shear stress and the velocity at various positions downstream of the film such that the correlations peak at zero time delay. $\times$, $x/\delta = 0.24$, $y/\delta = 0.05$; $+$, $x/\delta = 0.89$, $y/\delta = 0.25$; Δ , $x/\delta = 1.55$, $y/\delta = 0.50$; $\circ$, $x/\delta = 2.13$, $y/\delta = 0.75$.

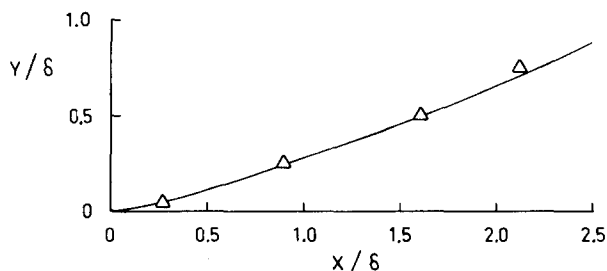


FIG. 3. Positions of the hot wires used to obtain the results in Fig. 2.

$x=0$ and the value of x for the maxima at $T=0$ ranges from $0.65 U_0$ at $y=0.05$ to $0.75 U_0$ at $y=0.75\delta$. (These velocities are more accurately given than might be expected from Figs. 1 and 2; they were obtained from the more accurately defined "enhanced" correlations to be discussed.) These velocities are all smaller than the local mean velocities which, if one point in the structure moves with the local mean velocity (see Sec. IV), suggests that the structure is not frozen but is, presumably, slowly growing and simultaneously very slowly rotating. This result and conclusion is not inconsistent with reported pressure fluctuation measurements¹ and space-time correlation results.¹³

Having concluded that an organized structure could account for these initial results more stringent tests of the hypothesis were now sought. Further progress appeared to require some technique for recognizing the particular pattern in the signals which could be associated with the structure and for the reasons given in the introduction this is the difficult and hazardous step. The approach adopted in this work is as follows. All pattern recognition is in essence a question of correlation, and it was argued that if the large structure accounted for the observed long-time wall-shear velocity correlations, then there should be short periods of time when the correlation is much greater than this long time correlation.

The output of the shear stress probe was therefore broken up into a large number (400) of short records each $T_s (= 7.5)$ milliseconds long. This length of time, i.e., $T_s U_0 / \delta^* = 50$ corresponds to about twice the characteristic time scale for the $y=0.25\delta$ correlation shown in Fig. 1. Each of these short records was then correlated with the corresponding record of the hot-wire output at $y/\delta = 0.25$ (the latter delayed by the time delay T), i.e.,

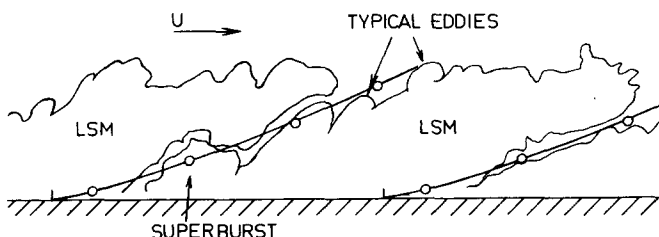


FIG. 4. Comparison of the wire positions required for correlations to peak at zero time delay (shown -o-o-) with a tracing of a smoke picture from Falco.¹⁹

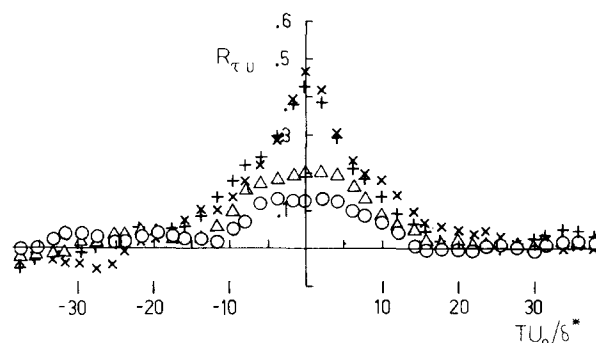


FIG. 5. "Enhanced" correlations obtained from an ensemble of various short time correlations taken at those times when the correlation between the wall shear stress and velocity at $y/\delta = 0.25$ is greater than 0.3 at zero time delay. The symbols are as in Fig. 2. Note the change in ordinate scale from Figs. 1 and 2.

$$R_n(T) = \frac{1}{T_s} \int_{t_n}^{t_n+T_s} \tau(t) u(t+T) dt.$$

Thus, a set of 400 short time correlations was obtained, the ensemble average of which was, of course, the $y = 0.25\delta$ curve in Fig. 2. These short time correlations were now compared with the ensemble average, and it was argued that if any supposed large structure was well represented in any 7.5 msec of output the short time correlation for this short record of data would be similar in shape but very much larger than the long time correlation. This was the basis of the pattern recognition. That is, from the set of 400, a subset (in fact, of 117) was obtained in which the criterion that the maximum value of the short time correlation be *twice* the long-time average (i.e., exceed 0.3) at zero time delay was satisfied. Obviously, the ensemble average of this subset of 117 had a maximum greater than twice the long-time average but apart from this greater amplitude the two correlations had the same general shape and the same time scale. (This will be clear from Fig. 5.)

A test of the organized structure hypothesis is now self-evident. For this subset of times, the correlation between the wall shear and velocity at 0.25δ is much greater than the long-time correlation and if this enhanced correlation results from a discrimination against those times when an organized structure is less evident, then for this *same* subset of times the correlations between the wall shear and the velocity at other points across the layer must also be larger than the long-time average. Figure 5 shows the result of this test, that is the ensemble average of the short-time correlations at other points based on the subset of times determined by the wire at 0.25δ . It is quite clear that an increased correlation at 0.25δ gives a correspondingly increased correlation throughout the layer, and this the authors suggest is significant evidence for the existence of some organized structure. (It should be added that this increased correlation is not due to a discrimination on the basis of amplitude). Although it made no essential difference in the results, in finally finding the subset of times of large short-time correlation at $y = 0.25\delta$, each short-time correlation was

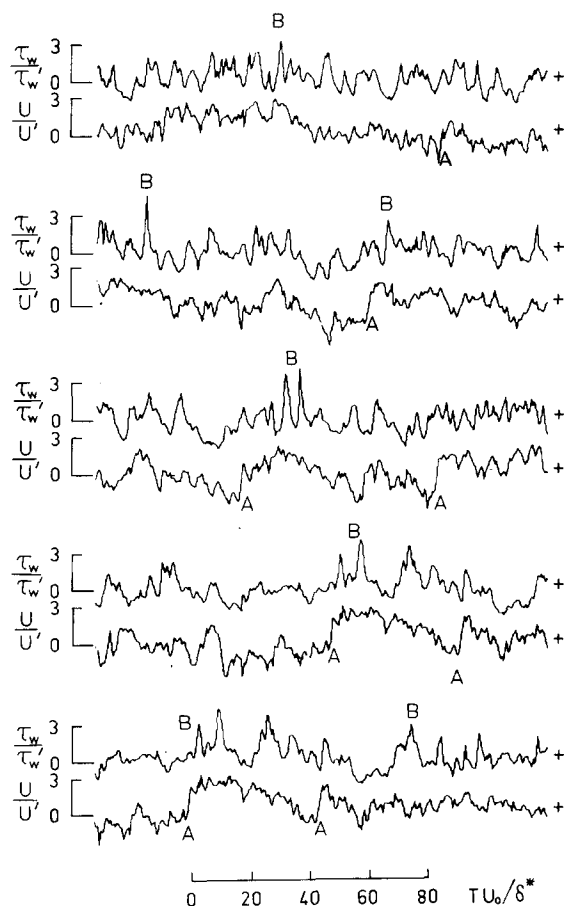


FIG. 6. Time records of the wall shear stress and velocity at $y/\delta = 0.25$ at those times when the detection criterion of Fig. 5 was satisfied.

nondimensionalized by the rms values of the signals during this short-time (thus discriminating entirely on the basis of correlation and not amplitude). For the subset of times of large correlation, the rms value of the signal taken over the whole subset of times was used to nondimensionalize the ensemble average of the corresponding short-time correlations. It is important to note, however, that these rms values obtained for this subset of times were all within 10% of the usual, long-time average rms values.)

We also note in passing that the criterion used to obtain the subset (i.e., the required value of the maximum correlation) was varied. Increasing this value to three and four times the long-time correlation reduced the percentage of records satisfying the criteria to 15% and 7%, respectively, but in every other way only substantiated the result. It is felt, however, that the large number of records in which the short-time correlation has a crudely similar shape, but very much greater correlation than the long-time average, is an indication that whatever accounts for this correlation is not in any sense an exceptional or rare event which might therefore make it insignificant in the mechanics of the flow, but rather it is a basic element of which the flow is composed.

Our attention was then turned toward finding some of

the characteristics of this apparent structure. Careful observation of the long-time signal records of all five probes led to little, but it is clear that the place to look for these characteristics is in the short records of data in which the short-time correlation is the same shape as the long-time correlation, but the maximum value is much larger than that of the long-time correlation. Figure 6 shows a sample of a number of these records selected by computer on this basis. The correlation between the wall shear and the velocity at 0.25δ is now overwhelmingly clear (by turbulence standards!). Several features in these signals are of interest. Firstly, the velocity trace frequently shows a relatively sharp "step" from a lower value to a higher value (marked by A in Fig. 6) followed by a slow decrease in velocity. That this is typical is supported by the skewness in the time derivative of the velocity. This feature is even more apparent at larger y/δ as is clear in Fig. 7 which shows the outputs of the five probes for a similar period. The characteristic sudden change in velocity may be seen to frequently extend across the full boundary layer. Laufer²⁰ shows an example of exactly this characteristic found in the output of an array of temperature probes in a slightly heated boundary layer. As will be discussed, it appears that this feature occurs on the "back" or trailing surface of the large structure. Laufer²⁰ arrived at the same conclusion.

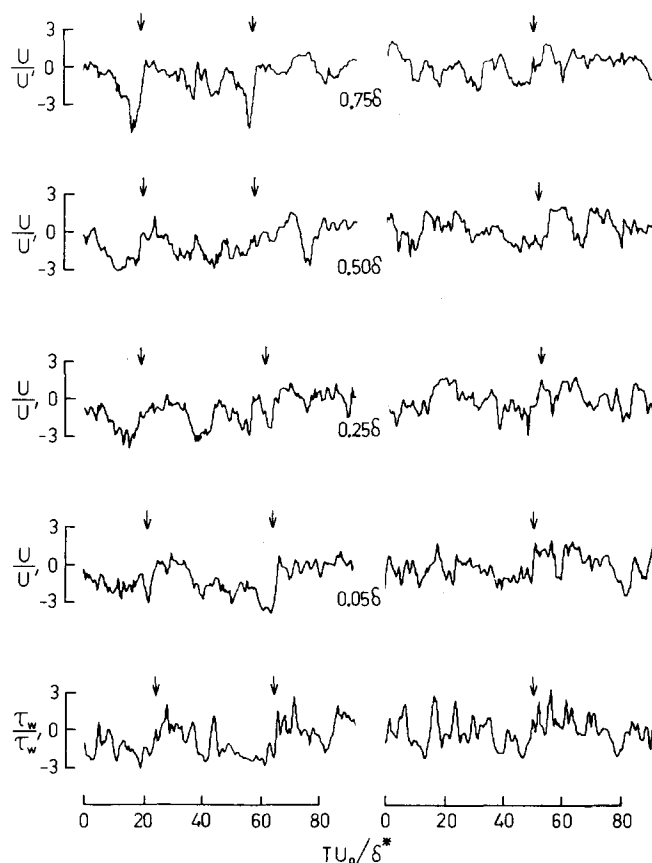


FIG. 7. Time records of the wall shear stress and velocity at $y/\delta = 0.05, 0.25, 0.50$, and 0.75 at those times when the detection criterion of Fig. 5 was satisfied.

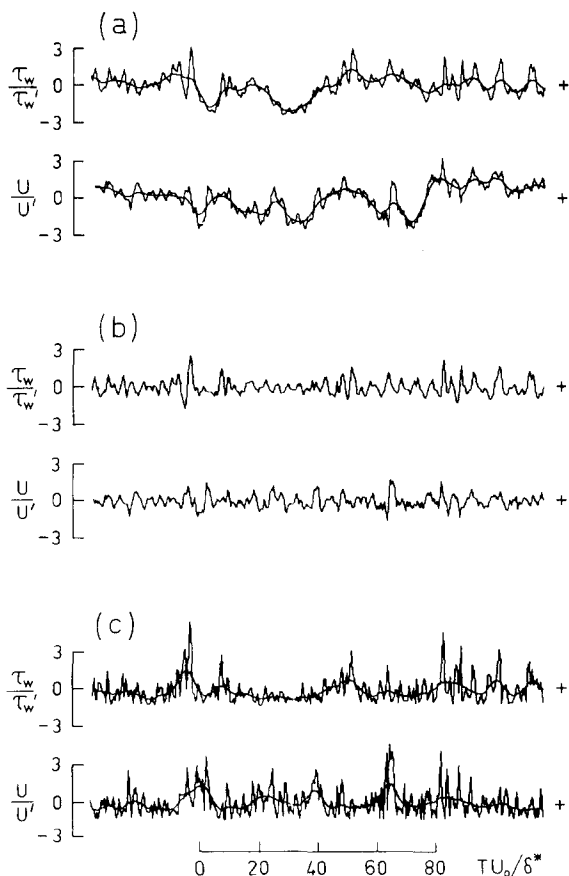


FIG. 8. Time records of the wall shear stress and velocity at $y/\delta=0.25$ when subjected to smoothing and rectifying. (a) Unfiltered signals with low pass filtered signal superimposed. (b) The high frequency signal obtained by taking the difference between the two signals in (a). (c) The rectified high frequency signal and the smoothed rectified high frequency signal.

A second feature in Fig. 6 which it seemed important to notice is the apparent large fluctuations in the wall shear (marked B) which frequently occur when the more slowly varying part of the wall shear is relatively high. A technique to quantify this is discussed in Sec. IIIB. The possible implication is that the slowly varying part of the wall shear results from the passage of the large structure and this characteristic high frequency fluctuation is part of the dynamics of the organized structure.

B. The wall shear fluctuations

The qualitative ideas mentioned here suggest that the wall shear signal be split into a slowly varying part and a high frequency part representing large scale and small scale motions. This was done, more or less intuitively, as follows: A zero-phase delay digital filter was used to smooth the wall shear (and velocity) signals. Figure 8 shows the original and smoothed curves superimposed. The frequency of the filter cutoff was varied to give "acceptable" smoothing and was finally set at a frequency of $f\delta^*/U_0=0.068$. [It may be noted that this frequency corresponds to roughly twice the bursting frequency ($TU_0/\delta^*\approx 30$) of Rao *et al.*,⁶] Figure 8 shows the remaining high frequency part of the signal

after the smoothed curve is subtracted from the original signal. Any attempt to mark the large amplitude regions of the high frequency signal and obtain some sort of intermittency function would clearly be problematical, and it seemed more appropriate to achieve the same end by simply rectifying the high frequency signal and then placing an envelope, more accurately, smoothing this rectified signal. Figure 8 shows the result (the nondimensional filter cutoff frequency was again 0.068). Qualitatively, this whole process clearly achieves the original aim, namely, to separate the slowly varying part of the signal and the regions of large amplitude high frequency fluctuations. The resulting low frequency and rectified high frequency smoothed signals (marked in Figs. 8(a) and (c)) were now correlated and the result for the wall shear is shown in Fig. 9. The time delay is in the sense of the smoothed rectified high frequencies at time t and the low frequency at time $t+T$. The correlation is evidently very large indeed. (The rms levels used to normalize the correlations are those of each smoothed signal.) It is interesting that the correlation peaks at a nonzero time delay implying that there is a particular phase in the low frequency wall fluctuation at which the high frequency activity is most likely to occur.

This same process was now applied to the velocity signals obtained at various distances from the wall and the results are shown in Fig. 10. These results reflect what could perhaps have been anticipated from Fig. 7, namely, the increasing appearance of the sharp step in velocity (discussed in Sec. IIIA) with distance from the wall. It is interesting to note that substantial correlations exist between the rectified high frequency content of the velocity signals even at $y=0.75\delta$ and the low frequency content of the wall shear. From the time delay in the peaks in this correlation, it is concluded that the sharp velocity step occurs on the "back" or trailing surface of the large structure. [That this technique did not generate a correlation when none existed was demonstrated by applying it to random noise and various pseudo turbulence signals (similarly skewed). In all cases a negligibly small correlation was obtained].

Without wishing to cloud the picture, there does seem some point in the following digression. Given the

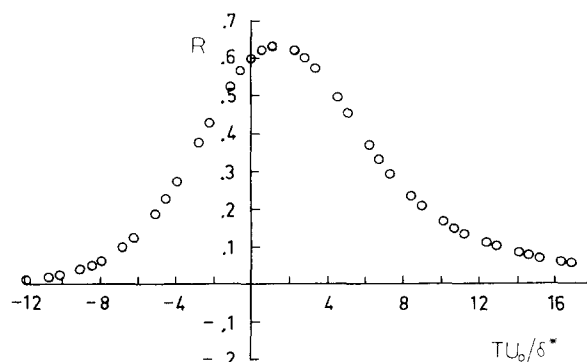


FIG. 9. Correlation between the low pass filtered wall shear stress and the smoothed, rectified high frequency component of the wall shear stress.

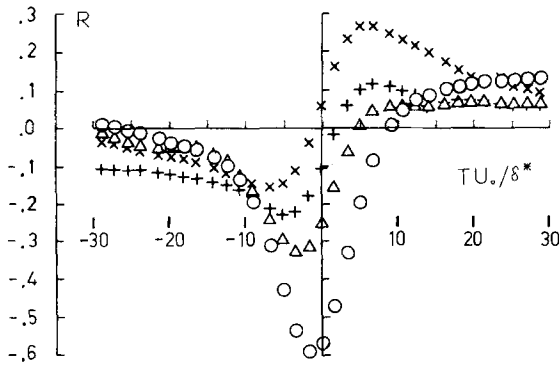


FIG. 10. Correlation between the low pass filtered velocity and the smoothed, rectified high frequency component. The symbols are as in Fig. 1.

correlation between low frequency and high frequency components of the wall shear it seemed appropriate to only correlate the low frequency (slowly varying) part of the wall shear [i. e., signal (a) in Fig. 8] with the similarly obtained low frequency component of the velocity signal at various distances from the wall. From the setting of the filter and the appearance of the signals it seemed that only the smoothed signals would make up the major part of the correlation between the two. This is indeed the case. Taking the correlation between the two smoothed signals and using the original rms values to nondimensionalize the correlation produced no apparent change to the correlation. It is then clear, however, that only the rms value of the smoothed signals should be used in correlating the two. The point of this is more than simply a trivial question of scaling, or a scandalous attempt to produce a correlation of 1. One is interested in the extent to which the large structure produces a correlated large scale motion in the boundary layer and this should be done without including the small-scale "hash" so to speak. (It contributes little or nothing to the correlation and it is sensible to not include it when nondimensionalizing the correlation). Thus, when the smoothed shear and velocity curves and the rms values for these smoothed curves are used to compute the results shown in Fig. 5, the results shown in Fig. 11 are obtained. A direct comparison can now be made between Figs. 2, 5, and 11. The extent to which the organized structure does produce a highly correlated large scale motion in the boundary layer can now clearly be assessed.

IV. DISCUSSION

All of these results taken together suggest that there is an organized structure in the boundary layer. The structure appears to be at an oblique angle to the wall and the most probable value of this angle from these measurements is 18° . From the time scale of the wall shear velocity correlations, the structure is of the order 2δ long at the wall. There seems strong evidence that a part of the dynamics of this organized structure is a relatively small scale large amplitude wall shear fluctuation which occurs just before the local maximum in the "slowly varying" part of the wall shear. This slowly varying part of the wall shear is itself caused

by the large structure and is on the same scale as the large structure. Alternatively, it could be said that the passage of the large structure gives rise to a characteristic response in the region near the wall and this response is observed as a high frequency large amplitude wall shear fluctuation.

It seems likely that this wall shear fluctuation is directly connected with the bursting phenomenon. In fact, the cycle of bursts and sweeps, the generation of new streaks and all of the present results can be shown to be consistent with the following model of the organized structure. For the present authors, the origin of this model was the suggestion by Coles and Barker⁸ that the turbulent spot may be the building block of a turbulent boundary layer. The ideas that led to the description of the burst cycle had their origin in recent work by Konrad and Brown.²¹ In this work, it has been found that the large structure in the mixing layer has its own internal instability at a particular Reynolds number and above this Reynolds number there is a very marked increase in small scale mixing and a quite apparent cellular structure in the basic large structure of the flow [see Brown and Roshko¹⁰ Fig. 8(b) for some indication of this]. The explanation for this instability offered by Konrad and Brown is that it is a result of a local rotational instability [$d(\Gamma^2)/dr < 0$] that is a continuously recurring condition due to the formation and amalgamation of large structures.

(Before discussing the present model we add that further experiments have been undertaken with several wall shear probes, with an array of hot wires spaced in the transverse direction and with crossed wires spaced across the layer. The final results of this work, particularly the streamline pattern are not yet at hand, but they do not appear to contradict the qualitative features of the model described.)

A schematic for the model of the organized structure in the boundary layer is shown in Fig. 12. In a coordinate system fixed in the large structure, that is a coordinate system in which the streamlines are changing only slowly, the organized structure appears as a

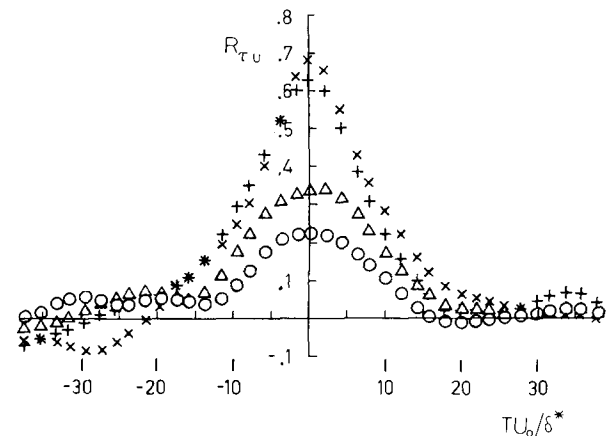


FIG. 11. Enhanced correlations that are similar to those in Fig. 5 except that low pass filtered signals have been used. The symbols are as in Fig. 2.

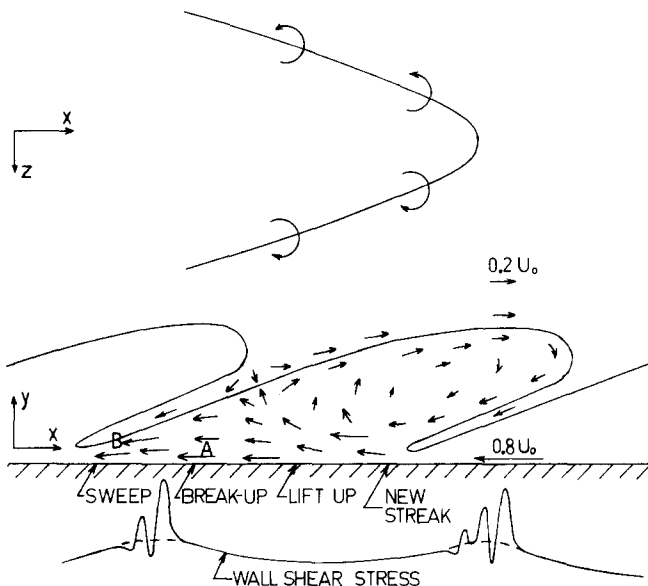


FIG. 12. Proposed flow pattern that might be seen by an observer moving at a speed of $0.8 U_0$, and the associated wall shear stress distribution.

horseshoe vortex which then gives rise to the velocity field sketched in Fig. 11 in the center-line plane of the vortex. It is expected from the present results (Sec. IIIA) and from pressure fluctuation measurements, that in this coordinate system the wall moves with a velocity of approximately $0.8 U_0$. The stagnation point is then on the rear surface of the large structure and on or near this surface it is readily believable that the sharp step in velocity discussed in Sec. IIIA occurs. Such a local shear layer is very likely to give rise to the "typical" eddies marked by Dr. Falco in the tracing of Fig. 4. The expected low frequency variation in wall shear due to the velocity field generated by this large structure is also shown in Fig. 11. Marked on this figure is the most probable position for the observed large fluctuations in shear stress and in addition the approximate locations of where the formation of a "new streak," "lift-up," and "break-up" have been observed^{22,23} in the "burst" cycle. These results may be seen to fit well in the context of the present organized structure hypothesis. In particular, attention is drawn to the region marked A where the stream lines are convex with respect to the surface and the region marked B where there is evidently a longitudinal straining motion. As discussed more quantitatively in Sec. V it is noted that in region A the mean flow vorticity and circulation are of opposite sign. For a steady flow this is the criterion given by Rayleigh²⁴ for an inviscid instability. For a viscous fluid it is expected that a characteristic Taylor number will exist for this region above which a cellular structure of Taylor or Görtler vortices will be generated. (That is a secondary flow similar to Taylor vortices between rotating cylinders²⁵ or Görtler vortices in a boundary layer on a concave wall.²⁶) This, it is suggested, is the origin of the longitudinal vortices or streaks in the wall region and at higher Reynolds numbers this basic instability as-

sists in the "breakup" phase of the burst cycle. (The authors understand that Coles²⁷ has independently considered this mechanism as part of the dynamics of the turbulent spot. As a further consequence of this model it is suggested that as this longitudinal vorticity arising from the instability is convected into region B the longitudinal straining motion will intensify this vorticity and return it toward the wall thereby assisting in the formation of a new streak. As a result of this, it is possible that very substantial streamwise correlations in the wall region could exist over distances of the order of δ , that is very much longer distances, at high Reynolds number, than characteristic wall scales. This too is in accord with experiment. It is also noteworthy that the model would account for the sensitivity of the turbulent boundary layer to radius of curvature of the wall, particularly to concave surfaces. (Care is required in attempting the Galilean transformation.)

Lastly, it is interesting to speculate whether the dramatic effect of drag reducing polymers is due to the ability of the polymer to resist the formation of this longitudinal vorticity in the "unstable" region in which it would otherwise be generated. Within the context of the present model it might be expected that for very small concentrations of polymer the large structure would itself not be greatly affected by the polymer, but the response of the wall region to the passage of the structure could be greatly modified due to this resistance. There would, correspondingly, be a marked reduction in Reynolds stress near the wall and a not very greatly changed external flow in the boundary layer.

A question which naturally arises is what is the origin of this large structure. It is quite plausible that it arises as a turbulent spot as Coles and Barker⁸ suggest and that the spots interact, amalgamate, etc., and form a relatively stable and not rapidly growing structure. It is likely, however, that such a structure will not be the building block of fully developed pipe flow and it is curious that the flow near the wall is so similar to that in a boundary layer. We note, however, that the flow near the wall requires only a slowly varying shear stress due to a passing larger scale motion to give rise to the instability described, and it is not difficult to imagine such a large scale motion in pipe or channel flow.

V. ESTIMATES FOR THE INSTABILITY IN THE ORGANIZED STRUCTURE MODEL

It is worth pursuing the discussion of Sec. IV in an endeavor to put the justification for the burst cycle, in terms of a Taylor-Görtler instability, on a more quantitative basis.

Assuming that the velocity near the wall depends on the local wall shear then in the coordinate system of approximately *steady* flow the velocity is

$$u = -U_c + U_\tau f(y U_\tau / \nu),$$

where U_c is the convection speed of the large structure. The stream function at a height δ is then given by

$$\psi = -U_c \delta + \nu \int_0^{\delta^*} f(y^*) dy^*,$$

where as usual $\delta^* = \delta U_\tau / \nu$. It follows that for any ψ if U_τ varies with x , then

$$\delta(x) = \frac{\nu}{U_\tau(x)} F\left(\frac{\psi}{\nu}, \frac{U_\tau(x)}{U_c}\right).$$

The radius of curvature R of the stream line at $y = \delta$ is given by

$$1/R = d^2\delta/dx^2.$$

For the boundary layer profile near the wall simple functions may be used to approximate $f(yU_\tau/\nu)$. As a quite accurate representation for ψ (within 3%) in the range $15 < \delta^* < 45$ we obtained

$$\psi + 95\nu = (13 U_\tau - U_c)\delta$$

from which we straightforwardly find, for small variations in U_τ , an expression for $1/R$. Using $U_c = 0.8 U_0$, $U_\tau = 0.035 U_0$, and $0.1 U_\tau$ for the amplitude of a sinusoidal variation in U_τ with $2\delta_0$ (where δ_0 is the boundary layer thickness) the wavelength of this variation, we obtain

$$1/R \approx \frac{4}{3} (\delta/\delta_0^2).$$

The rotational instability criterion found by Taylor²⁵ can be written in the form

$$\bar{\Omega} \bar{\zeta} \leq -\nu^2/2L^4,$$

where $\bar{\Omega}$ is the characteristic angular velocity, $\bar{\zeta}$ is the local vorticity, and L is a characteristic viscous dissipation scale for the secondary motion. This result follows directly from orders of magnitude estimates in the linearized vorticity equations and has been expressed in this way by Coles.²⁸ In using this result he has based estimates for L on the values of L in flows of known instability (i.e., the flow between rotating cylinders). In applying this inequality in the present context we take the angular velocity to be u/R , the vorticity to be $U_\tau^2/\nu f'(y^*)$ and, based on Coles results, L equal to $2\delta/5$. With this estimate for R , the known function $f(y^*)$, and typical values for U_τ/U_0 and δ^*/δ_0 the inequality reduces, approximately, to

$$\delta^* \geq 1.2 (U_0 \delta^*/\nu)^{2/5}$$

which for the present experiment requires $\delta^* \sim 53$. The wall region is thus unstable by an inviscid Rayleigh criterion and for a viscous fluid is also unstable with the center of the instability estimated at a height of $\delta^* \sim 50$. This is, of course, nothing more than an order of magnitude estimate and a demonstration of the plausibility of the qualitative discussion. Changing the ratio between L and δ will have an almost linear effect on the estimate of δ . Reynolds number has a weak effect, although it is stronger than might have been hoped for. Nevertheless, the authors feel that these estimates support the model and are not inconsistent with a measured streak spacing of 100 wall units.

VI. CONCLUSION

The results presented point very strongly to the presence of an organized structure in the boundary layer. This structure is on the scale of the full boundary layer. Previous correlations in the outer part of the boundary layer are consistent with this structure and the bursting process near the wall, as inferred in this study from measurements of high frequency wall shear fluctuations, is shown by correlation techniques to be a direct response of the region near the wall to the large scale motion of the organized structure. The techniques employed to recognize organized structure and to then conditionally sample and improve correlations for the purpose of clarifying the properties of this structure are thought to be more widely applicable. A model for the organized structure which is consistent with the present results is presented. While speculative at this stage, it attempts to account for the bursting phenomenon near the wall by demonstrating that the slowly varying wall shear imposed by the large structure gives rise to sufficient streamline curvature, in a steady flow coordinate system, to produce a Taylor-Görtler type of instability. It is shown that this instability may be centered at a δ^* of the order of 50 in the present experiments so that a streak spacing of the order of 100 wall units is not inconsistent with the model.

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